GNBF5020

Midterm Drill Pack

期中考試操練手冊

A complete drill pack for the 20 + 10 midterm: the exam format, the five question types he named, 70 single-best-answer MCQs with distractor analysis, and two worked scenario templates.

GNBF5020 · Introduction to Molecular Biology and Genetics
Dr. Tony Wong · Lectures 1-5, 12 Sep - 17 Oct 2026

> **[ENGLISH ONLY]** The exam must be answered in English. Dr. Wong said this three separate times, and once he said he gave zero marks for it. This document is in English so you can revise directly in the language you will be graded in. The Chinese companion exists only for explanation; gene, protein, pathway, method and drug names stay in English in both versions.

Contents

GNBF5020 — Midterm Drill Pack

A complete drill pack for the 20 + 10 midterm

1. The exam, exactly as he describes it

The format conflict, and which one to plan for

Timing plan for 90 minutes

2. The four rules for the short questions
3. The five question types he named as exam-type

3.1 Reading a Kaplan–Meier curve — Lecture 1

3.2 Genetic counselling from a medical report — Lecture 1

3.3 A defect in NHEJ gives you SCID — Lecture 4

3.4 Reading a mitochondrial pedigree — Lecture 2

3.5 Reading the lac operon logic correctly — Lecture 5

4. MCQ drill bank — 60 single-best-answer questions

Lecture 1 — Introduction to Biomolecules

Lecture 2 — Basic Cell Biology

Lecture 3 — DNA and Chromatin Structure

Lecture 4 — DNA Replication, Repair and Recombination

Lecture 5 — Control of Gene Expression

5. Two worked scenario templates

Template A — Pedigree / genetic counselling

Template B — Survival data / Kaplan–Meier

6. Common traps to avoid
7. Seven-day plan to the midterm
8. Final self-test

GNBF5020 — Midterm Drill Pack

A complete drill pack for the 20 + 10 midterm

Course: GNBF5020 Introduction to Molecular Biology and Genetics · Dr. Tony Wong · Lectures 1–5, 12 Sep – 17 Oct 2026 **Format:** 20 multiple-choice + choose 2 of 3 short questions = 30 marks · 90 minutes · English only

> ***How to use this pack.*** *The other documents in this kit teach you the content. This pack trains you on the exam format: the shape of the paper, the question types he has said out loud he will ask, and a bank of single-best-answer MCQs to drill under time pressure. Work through it in order.*

1. The exam, exactly as he describes it

Item	Specification	Where he said it
Midterm weight	**30%** of the final grade	Lecture 1, Slide 2
Final exam	40% of the final grade	Lecture 1, Slide 2
Two assignments	30% (“super-easy”)	Lecture 1, Slide 2
Part A	**20 multiple-choice questions, single best option only**	Lecture 5 (later statement)
Part B	**Choose 2 of 3 short questions** (data analysis / scenario)	Lecture 5
Total	**30 marks in 90 minutes**	Lecture 5
Language	**English only — zero marks for Chinese answers**	Lectures 1, 4, 5

The format conflict, and which one to plan for

Lecture	Date	What he says
Lecture 4	10 Oct 2026	“about 25 questions, one mark each … two questions, you choose only one short question to answer”
Lecture 5	**17 Oct 2026**	**“we have 20 multiple choice questions, and two out of three short question … 20 marks … short question is account for 10 marks … 1.5 hours”**

Lecture 5 is the later statement and it is written on his own title slide, so **plan for 20 MCQ + 2 of 3 short questions**. The ratio and the strategy are identical either way.

> ***The strategy he states himself:*** *“multiple choice is for you to pass the exam. But if you want to get a full mark or a better result, you’d better perform well for the short questions, which emphasize on data analysis and also the*

> *scenarios questions." — Lecture 1*

So: **20 MCQs are the floor. The 10 short-question marks are the difference.**

Timing plan for 90 minutes

Minutes	Task	Why
0–5	Read the paper; identify which 2 of the 3 short questions you will attempt	You choose, so choose early
5–35	All 20 MCQs, ~90 seconds each	Fast marks; do not linger
35–80	The 2 short questions you chose, ~22 minutes each	This is where the grade is
80–88	Check MCQs you were unsure about	Cheap marks on a re-read
88–90	Check you wrote full sentences, in English	Zero marks are possible here

---

2. The four rules for the short questions

These come from what he said in the lectures, and each one has cost marks in previous years.

1. **Write in English, always.** He said it three times. He said he gave **zero** marks last year for Chinese answers. Simple short English sentences score better than no English.
2. **Write full sentences.** His advice is to *defend your reasoning*. **A page of bullet points is not an answer.** A bullet point is a note; an answer is a sentence that argues.
3. **Keep the technical terms in English inside those sentences.** Write *"the KRAS mutation means cetuximab will not work"*, not a Chinese sentence with one English word in it.
4. **Show the chain, not the conclusion.** For every scenario question, mark the steps. A conclusion without the chain gets partial credit; a chain with a conclusion gets full credit.

---

3. The five question types he named as exam-type

These are the highest-confidence items in the course. He said in his own words that these are the question types he puts in exams. **Learn the chain in each one, not just the answer.**

3.1 Reading a Kaplan–Meier curve — Lecture 1

> *"And I ask this single question every, every time when I talk about the KM plot … And only a single student give me a good answer last year." — Lecture 1*

What the axes mean - **x-axis:** time (months) - **y-axis:** probability of survival, or of progression-free survival - Curve **starts at 1.0** and falls in **steps**

How to read each element

Feature	Meaning
Flat segment	No events occurred in that interval

Feature	Meaning
Downward step	An event occurred (death, or progression)
Small tick mark on the curve	A **censored** patient — lost to follow-up, or still alive at the cut-off
Steep early drop	The gene is associated with **early involvement** and a poor prognosis
Late drop / long plateau	Patients do well early; the event is associated with **late involvement**
Median survival	Where the curve crosses 0.5

The chain to draw in your answer

```
1. Read the axes          → what is x, what is y, what is the unit
2. Identify the groups    → which curve is which treatment / which genotype
3. Find the separation    → where and how far apart do the curves diverge?
4. Interpret timing       → early separation = early involvement; late = late involvement
5. Name the clinical use  → which biomarker decision does this support?
6. State the caveat       → censored ticks mean the estimate is conservative
```

The sentence to write: *"The curve for group A falls steeply within the first six months, whereas group B has a long plateau before any step, so the gene under investigation is associated with early involvement and a poorer prognosis in group A."*

3.2 Genetic counselling from a medical report — Lecture 1

> *"if you're a genetic consultant and I give you a medical report, how you would suggest the client whether she or him will get a baby with a certain genetic diseases?" — Lecture 1*

The chain to draw

```
1. Identify the inheritance pattern from the pedigree
      · autosomal dominant   → vertical, every generation, male→male transmission possible
      · autosomal recessive  → horizontal, skips generations, siblings affected, parents unaffected
      · X-linked recessive   → mostly males, no male→male transmission
      · mitochondrial        → affected mother passes to ALL children; affected father passes to NONE
2. Work out the mode of inheritance FIRST
3. Then compute the risk for THIS couple
      · autosomal recessive: risk = carrier probability of each parent × 1/4 per pregnancy
      · X-linked recessive: risk depends on the carrier state of the mother
      · mitochondrial: if the mother is affected, all children inherit; the father is irrelevant
4. State the probability PER PREGNANCY (not per child)
5. Say what you would do next — genetic testing, or IVF with preimplantation testing
```

The sentence to write: *"The pedigree is most consistent with autosomal recessive inheritance, because affected siblings are born to two unaffected parents and the trait skips a generation. If both parents are obligate carriers, the risk to each pregnancy is one in four. I would recommend testing both parents to confirm carrier status, and, if they wish to avoid transmission, preimplantation genetic testing with IVF."*

3.3 A defect in NHEJ gives you SCID — Lecture 4

> *"if I ask you, if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency … because he cannot generate a sufficient amount of the antibody pool." — Lecture 4*

The chain to draw

```
NHEJ  →  V(D)J recombination  →  assembles V, D and J gene segments
      →  antibody diversity in B-cells + T-cell receptor diversity in T-cells
      →  the repertoire cannot be built
      →  SCID (severe combined immunodeficiency)
```

The numbers he gave: V ≈ 39 copies, D = 27 copies, J = 6 copies.

The sentence to write: *"NHEJ is co-opted by V(D)J recombination to join the V, D and J gene segments, and this is what generates antibody diversity in B-cells and T-cell receptor diversity in T-cells. A defect in NHEJ therefore prevents the repertoire from being built at all, and the clinical outcome is severe combined immunodeficiency."*

3.4 Reading a mitochondrial pedigree — Lecture 2

> *"Not only the mutation within the mitochondrial DNA can cause the mitochondrial disease. Some mitochondrial disease can actually result from the mutation within the nuclear DNA … different mutation can cause the different inheritance patterns." — Lecture 2*

The three hallmarks of mitochondrial (maternal) inheritance 1. An **affected mother passes the condition to all** of her children 2. An **affected father passes it to none** 3. **Only females transmit** it onward

The trap: the *disease* can be caused by a mutation in **nuclear** DNA too, and **then** it inherits in a Mendelian way. Do not assume mitochondrial inheritance from the pedigree alone — ask *which* gene.

The chain to draw

```
1. Look at who transmits and who does not
2. Maternal pattern?  → mitochondrial DNA mutation
3. Mendelian pattern? → nuclear DNA mutation (so NOT mitochondrial inheritance)
4. Name the gene / type
5. Give the recurrence risk
```

The sentence to write: *"The pedigree shows transmission through affected mothers to all their children and none from affected fathers, which is the maternal pattern of mitochondrial inheritance and therefore points to a mutation in mitochondrial DNA. Because the condition can also be caused by nuclear-gene mutations, I would confirm the genotype before giving the final recurrence risk."*

3.5 Reading the lac operon logic correctly — Lecture 5

> *Tier 2 — he says you must understand it. It is also scenario case 16 in the case-study bank.*

The two conditions for strong transcription — both must be true at once

Condition	What it does	Molecule
Lactose present	Allolactose binds the Lac repressor and knocks it off the operator	**Allolactose** (made by lacZ-encoded β-galactosidase — *not* lactose itself)

Condition	What it does	Molecule
Glucose low	cAMP rises; the cAMP–CAP complex binds the promoter and recruits the copying machinery	**cAMP–CAP**

> ***Naming only "lactose" is the classic wrong answer.*** *It scores half marks at best.*

The three genes: lacZ (β-galactosidase), **lacY** (permease), **lacA** (transacetylase) — copied together as one polycistronic message.

The trp operon is the mirror image: tryptophan binds the Trp repressor as a **corepressor** and switches off tryptophan biosynthesis. Its five genes: **trpE, trpD, trpC, trpB, trpA**.

IPTG is the artificial, non-metabolised inducer used in *E. coli* recombinant systems. Optimal concentration on his slide: **100 μM** (range tested 0–1000 μM).

The sentence to write: *"Strong transcription of the lac operon needs two conditions at once: lactose must be present so that allolactose binds the Lac repressor and removes it from the operator, and glucose must be low so that cAMP rises and the cAMP–CAP complex binds the promoter. Either condition alone gives only weak transcription."*

---

4. MCQ drill bank — 60 single-best-answer questions

Drill these under time: **90 seconds each**. If you cannot decide, mark it and move on; you have 20 marks of floor to protect.

Each answer lists the **distractor** that catches people, because he has said the MCQs are where students lose marks by "trying to select multiple options."

Lecture 1 — Introduction to Biomolecules

Q1. Which statement about biomolecules is correct? - A. All four classes are polymers built from monomers - B. Lipids are polymers of fatty acids joined by peptide bonds - C. Carbohydrates, lipids, proteins and nucleic acids are the four classes of biomolecule - D. Proteins are the only class that form by dehydration condensation - E. Nucleic acids are built from amino acids

▶ **Answer**

Q2. Which bond links the two nucleotides of a phosphodiester linkage? - A. A hydrogen bond between complementary bases - B. A peptide bond between the 3′-OH and the 5′-phosphate - C. A glycosidic bond between two sugars - D. A disulfide bridge between two cysteines - E. A π bond within the base

▶ **Answer**

Q3. Why does DNA contain thymine rather than uracil? - A. Thymine pairs with guanine - B. Uracil is unstable at physiological pH - C. Deamination of cytosine produces uracil, which is detectable and repairable; deamination of 5-methylcytosine produces thymine, which is not - D. Thymine has three hydrogen bonds - E. RNA is single-stranded and cannot contain uracil

▶ **Answer**

Q4. The primary level of protein structure is: - A. The α-helix and β-sheet - B. The sequence of amino acids joined by peptide bonds - C. The three-dimensional fold of the whole chain - D. The association of several subunits - E. Covalent bonds between distant residues

▶ **Answer**

Q5. A molecule that changes shape — and therefore its activity — when something binds at a site other than its active site is showing: - A. Denaturation - B. Allostery - C. Hydrolysis - D. Isomerisation - E. Saturating kinetics

▶ **Answer**

Q6. Which of these is *not* a function of lipids? - A. Energy storage - B. Membrane structure - C. Insulation - D. Catalysis - E. Signalling

▶ **Answer**

Q7. Alpha-helices and beta-sheets are examples of: - A. Primary structure - B. Secondary structure - C. Tertiary structure - D. Quaternary structure - E. Nucleotide sequence

▶ **Answer**

Q8. The bond that stabilises the tertiary structure of a protein by linking two cysteine residues is: - A. An ionic bond - B. A hydrogen bond - C. A peptide bond - D. A disulfide bond - E. A phosphodiester bond

▶ **Answer**

Q9. Which of the following is a monosaccharide? - A. Cellulose - B. Starch - C. Glucose - D. Sucrose - E. A nucleotide

▶ **Answer**

Q10. In a nucleotide, the nitrogenous base is attached to: - A. The 1′ carbon of the sugar - B. The 3′ carbon of the sugar - C. The 5′ carbon of the sugar - D. The phosphate group - E. The 3′ hydroxyl of the adjacent nucleotide

▶ **Answer**

Lecture 2 — Basic Cell Biology

Q11. Which structure is *not* found in a prokaryotic cell? - A. Ribosomes - B. A plasma membrane - C. A mitochondrion - D. Circular DNA - E. A nucleoid region

▶ **Answer**

Q12. A red blood cell placed in pure distilled water will: - A. Remain unchanged - B. Shrink - C. Burst (lyse) - D. Become crenated - E. Lose its nucleus only

▶ **Answer**

Q13. A red blood cell placed in a hypertonic solution becomes: - A. Hemolysed - B. Crenated (spiky and shrunken) - C. Larger and spherical - D. Converted to a reticulocyte - E. Unaffected

▶ **Answer**

Q14. Transport across the membrane that requires ATP is: - A. Simple diffusion - B. Facilitated diffusion - C. Osmosis - D. Active transport - E. Filtration

▶ **Answer**

Q15. The rough endoplasmic reticulum is studded with ribosomes because it is the site of: - A. Lipid synthesis - B. Protein synthesis for secretion and membranes - C. ATP production - D. DNA replication - E. Lysosomal degradation

▶ **Answer**

Q16. Mitochondria and chloroplasts are thought to have originated from: - A. Free-living bacteria engulfed by an ancestral cell - B. Folding of the plasma membrane - C. Division of the Golgi apparatus - D. Invagination of the nuclear envelope - E. Endocytosis of a large vacuole

▶ **Answer**

Q17. In a pedigree, an affected mother passes the trait to all her children and an affected father passes it to none. The inheritance is: - A. Autosomal dominant - B. Autosomal recessive - C. X-linked recessive - D. Mitochondrial - E. Y-linked

▶ **Answer**

Q18. Which phase of the cell cycle is DNA replicated? - A. G1 - B. S - C. G2 - D. M - E. Cytokinesis

▶ **Answer**

Q19. During mitosis, the sister chromatids separate and move to opposite poles in: - A. Prophase - B. Metaphase - C. Anaphase - D. Telophase - E. Prometaphase

▶ **Answer**

Q20. Crossing over occurs during: - A. Prophase I of meiosis - B. Metaphase I of meiosis - C. Anaphase I of meiosis - D. Mitosis - E. S phase

▶ **Answer**

Lecture 3 — DNA and Chromatin Structure

Q21. The two strands of a DNA double helix are: - A. Complementary and antiparallel - B. Complementary and parallel - C. Identical and antiparallel - D. Identical and parallel - E. Non-complementary and antiparallel

▶ **Answer**

Q22. Which statement about the major groove is correct? - A. It is wider than the minor groove - B. It is wider than the minor groove and is where many proteins read DNA sequence - C. It contains the sugar-phosphate backbone - D. It has no hydrogen bonds - E. It is found only in RNA

▶ **Answer**

Q23. A nucleosome consists of: - A. DNA wrapped around a histone octamer - B. RNA wrapped around a polymerase - C. A histone octamer wrapped around RNA - D. Free DNA with no proteins - E. A single H1 histone with 146 bp of DNA

▶ **Answer**

Q24. Which histone modification is most associated with **active** transcription? - A. H3K9me3 - B. H3K27me3 - C. H3K4me3 - D. H3K9me2 - E. DNA methylation at CpG islands

▶ **Answer**

Q25. Heterochromatin differs from euchromatin in that it is: - A. Less condensed and more transcriptionally active - B. More condensed and less transcriptionally active - C. Identical, but found only in the nucleus - D. Enriched in acetylated histones - E. The form found only during S phase

▶ **Answer**

Q26. The 5′ end of a eukaryotic linear chromosome is protected by: - A. A centromere - B. A telomere - C. An origin of replication - D. A satellite DNA block - E. A nucleosome

▶ **Answer**

Q27. The centromere of a chromosome is the region where: - A. Replication begins - B. Telomerase binds - C. Spindle microtubules attach via the kinetochore - D. RNA polymerase transcribes - E. Histone H1 is removed

▶ **Answer**

Q28. Which pair of bases forms **three** hydrogen bonds? - A. A–T - B. G–C - C. A–G - D. U–A - E. T–G

▶ **Answer**

Q29. In which cellular compartment is DNA replication carried out in eukaryotes? - A. The cytoplasm - B. The nucleus, during S phase - C. The mitochondrion only - D. The rough ER - E. The nucleolus

▶ **Answer**

Q30. Which of these is **not** a level of DNA packaging in eukaryotes? - A. Nucleosome - B. 30-nm fibre - C. Looped domains attached to a scaffold - D. Chromatid - E. Telomere

▶ **Answer**

Q31. Base stacking contributes to DNA stability mainly through: - A. Covalent bonding of the backbone - B. Hydrophobic interactions between adjacent base pairs - C. Ionic bonds with magnesium - D. Phosphodiester linkages - E. Disulfide bridges

▶ **Answer**

Q32. A “purine” is: - A. A base with one ring - B. A base with two fused rings - C. Any base that pairs with two others - D. A sugar-phosphate unit - E. A type of histone

▶ **Answer**

Lecture 4 — DNA Replication, Repair and Recombination

Q33. Meselson and Stahl’s result supports: - A. Conservative replication - B. Dispersive replication - C. Semi-conservative replication - D. Semi-discontinuous replication only - E. Rolling-circle replication

▶ **Answer**

Q34. Which enzyme relieves the torsional stress that builds up ahead of the replication fork? - A. Helicase - B. Topoisomerase - C. Primase - D. Ligase - E. Telomerase

▶ **Answer**

Q35. Why can DNA polymerase not start a strand from nothing? - A. It lacks the energy to open the double helix - B. It can only add nucleotides to an existing 3′-OH, so a primer is required - C. It cannot bind the template without ATP - D. It is inhibited by single-strand binding proteins - E. It can only add to the 5′ end

▶ **Answer**

Q36. On the lagging strand, the discontinuously made fragments are called: - A. Okazaki fragments - B. DnaA boxes - C. Telomeres - D. Centromeres - E. Exons

▶ **Answer**

Q37. Telomerase is: - A. A DNA polymerase that copies the lagging strand - B. A reverse transcriptase that uses its own RNA as a template - C. A helicase - D. A repair endonuclease - E. A topoisomerase

▶ **Answer**

Q38. Telomerase activity is found in stem cells and germ cells, and in: - A. Most somatic cells - B. About 90% of cancer cells - C. No human cells - D. Only mature red blood cells - E. Neurons only

▶ **Answer**

Q39. Which pathway removes a bulky lesion such as a UV-induced pyrimidine dimer? - A. Base excision repair - B. Nucleotide excision repair - C. Mismatch repair - D. Direct reversal by photolyase - E. Homologous recombination

▶ **Answer**

Q40. Which pathway fixes replication errors that escaped polymerase proofreading? - A. Base excision repair - B. Nucleotide excision repair - C. Mismatch repair - D. Fanconi anaemia pathway - E. Apoptosis

▶ **Answer**

Q41. A defect in nucleotide excision repair causes: - A. Xeroderma pigmentosum - B. Lynch syndrome - C. Hereditary breast and ovarian cancer - D. Cystic fibrosis - E. Sickle cell anaemia

▶ **Answer**

Q42. The genes BRCA1 and BRCA2 function in: - A. Mismatch repair - B. Base excision repair - C. Homologous recombination - D. Nucleotide excision repair - E. Non-homologous end joining

▶ **Answer**

Q43. Which statement about non-homologous end joining is correct? - A. It requires a sister chromatid as a template and is accurate - B. It joins broken ends without a template and may lose or gain bases - C. It excises a patch of

nucleotides around the lesion - D. It requires photolyase - E. It only functions during S phase

▶ **Answer**

Q44. A defect in NHEJ causes severe combined immunodeficiency because the cell cannot: - A. Synthesize DNA - B. Perform V(D)J recombination and so build antibody and T-cell receptor diversity - C. Replicate the mitochondrial genome - D. Perform base excision repair - E. Translate mRNA

▶ **Answer**

Q45. Homologous recombination during meiosis produces: - A. Sister chromatids - B. Non-sister recombinant chromatids - C. Identical daughter cells - D. Polyploid cells - E. Acrocentric chromosomes

▶ **Answer**

Q46. A transposon is best described as: - A. A protein that moves a chromosome to the nucleus - B. A mobile DNA element that moves its position within the genome - C. A type of spindle microtubule - D. A repair enzyme that removes thymine dimers - E. A region of a chromosome where the centromere forms

▶ **Answer**

Q47. Which of these mutagens is an alkylating agent that adds an ethyl group to a DNA base and commonly produces G→A transitions? - A. Ethidium bromide - B. Ethyl methanesulphonate (EMS) - C. Ultraviolet light - D. 5-bromouracil - E. Aflatoxin B1

▶ **Answer**

Q48. Which of these is **not** a spontaneous (endogenous) source of DNA damage? - A. Tautomerisation of bases - B. Depurination - C. Deamination of cytosine to uracil - D. Ultraviolet radiation - E. Oxidative damage by reactive oxygen species

▶ **Answer**

Q49. Direct reversal repair of a UV-induced pyrimidine dimer is carried out by: - A. Photolyase, using visible light energy - B. UvrABC, using ATP - C. DNA polymerase III - D. Ligase - E. Helicase II

▶ **Answer**

Q50. A cell with a defective DNA ligase would most obviously fail to: - A. Unwind the helix - B. Join Okazaki fragments on the lagging strand - C. Synthesize RNA primers - D. Remove uracil from DNA - E. Replicate the mitochondrial genome

▶ **Answer**

Q51. Which of these best distinguishes homologous recombination from NHEJ? - A. HR happens only in bacteria - B. HR uses a homologous template; NHEJ does not - C. NHEJ requires ATP and HR does not - D. HR occurs only during S phase - E. NHEJ is the only pathway that can repair double-strand breaks

▶ **Answer**

Q52. A thymine dimer distorts the DNA helix. Which pathway is designed to remove it? - A. Mismatch repair - B. Base excision repair - C. Nucleotide excision repair - D. Direct reversal by MGMT - E. Fanconi anaemia pathway

▶ **Answer**

Lecture 5 — Control of Gene Expression

Q53. The lac operon is switched on *strongly* only when: - A. Lactose is present - B. Glucose is low - C. Both lactose is present and glucose is low - D. Both lactose is absent and glucose is high - E. Tryptophan is scarce

▶ **Answer**

Q54. The molecule that inactivates the Lac repressor is: - A. Lactose itself - B. Glucose - C. Allolactose - D. cAMP - E. IPTG only in wild type

▶ **Answer**

Q55. The cAMP–CAP complex helps RNA polymerase by: - A. Binding the operator and blocking transcription - B. Binding the promoter and recruiting the transcription machinery - C. Degrading the operator - D. Converting glucose into lactose - E. Methylating the promoter

▶ **Answer**

Q56. The three structural genes of the lac operon are: - A. lacZ, lacY, lacA - B. trpE, trpD, trpC - C. araA, araB, araC - D. cAMP, CAP, and the repressor - E. p53, Rb and APC

▶ **Answer**

Q57. In the trp operon, tryptophan acts as: - A. An inducer that activates transcription - B. A corepressor that activates the repressor - C. An allosteric activator of CAP - D. A substrate of β-galactosidase - E. A transcriptional activator

▶ **Answer**

Q58. The five genes of the tryptophan operon are: - A. lacZ, lacY, lacA, lacI, lacO - B. trpE, trpD, trpC, trpB, trpA - C. V, D, J, RAG1, RAG2 - D. TP53, RB1, APC, KRAS, BRAF - E. p16, p21, p27, p53, Rb

▶ **Answer**

Q59. A **cis**-acting element is: - A. A diffusible protein that binds a distant gene - B. DNA sequence on the same DNA molecule that regulates the gene next to it - C. A transcription factor made in another cell - D. A microRNA - E. A histone mark

▶ **Answer**

Q60. The clearest example of trans-regulation is: - A. An enhancer on chromosome 13 activating a gene on chromosome 13 - B. A repressor encoded on chromosome 1 acting on a target gene on chromosome 13 - C. A promoter on the same chromosome as its gene - D. An operator next to the structural gene - E. A poly-A signal in the 3′ UTR

▶ **Answer**

Q61. Alternative splicing results in: - A. One gene producing several different proteins - B. Several genes producing one protein - C. Loss of all introns - D. Failure of transcription - E. Hyper-methylation of the promoter

▶ **Answer**

Q62. An RNA longer than 200 nucleotides that makes no protein is called: - A. A microRNA - B. A long non-coding RNA (lncRNA) - C. A small nuclear RNA - D. A tRNA - E. An siRNA

▶ **Answer**

Q63. A microRNA works by: - A. Binding a seed sequence in the target mRNA by partial base-pairing to suppress its translation - B. Binding the promoter and blocking RNA polymerase - C. Methylating the target gene - D. Cutting the DNA of the target gene - E. Recruiting histone deacetylase to the promoter only

▶ **Answer**

Q64. In bisulfite sequencing, unmethylated cytosine is: - A. Left unchanged - B. Converted to uracil, while methylated cytosine is left alone - C. Converted to thymine, while methylated cytosine is left alone - D. Converted to 5-methylcytosine - E. Degraded completely

▶ **Answer**

Q65. Which of these is a *trans*-acting epigenetic mechanism? - A. Promoter CpG methylation - B. Histone acetylation - C. Histone deacetylation - D. A microRNA - E. DNA methylation

▶ **Answer**

Q66. Which statement about cellular memory is correct? - A. A cell keeps a switch on long after the original signal has gone - B. A cell permanently loses the ability to change - C. It requires continuous signalling from outside the cell - D. It only occurs in prokaryotes - E. It is caused by DNA replication

▶ **Answer**

Q67. Which of these would be the expected effect of **hypomethylation** of a tumour suppressor promoter? - A. Increased expression of the tumour suppressor - B. Silencing of the tumour suppressor - C. No change in transcription - D. Increased RNA editing - E. Enhanced mismatch repair

▶ **Answer**

Q68. In CHO cells used for producing a therapeutic protein, Dr. Wong said you must: - A. Introduce a nuclear localisation signal - B. Perform codon optimisation - C. Use a prokaryotic vector - D. Add a poly-A tail to the genomic DNA - E. Delete the introns

▶ **Answer**

Q69. IPTG is used in recombinant *E. coli* systems because it: - A. Is a sugar that the cell can metabolise for energy - B. Is an artificial inducer that is not metabolised, so it switches on the lac system without being consumed - C. Directly binds RNA polymerase - D. Degrades lactose - E. Acts as a corepressor for the trp operon

▶ **Answer**

Q70. The NF-κB cytokine response proceeds in the order: - A. IFN-γ → IL-1β → TNF-α → IL-2 - B. TNF-α → IL-1β → IFN-γ → IL-2 and IL-15 - C. IL-2 → TNF-α → IFN-γ → IL-1β - D. TNF-α → IL-2 → IL-1β → IFN-γ - E. All four appear simultaneously

▶ **Answer**

---

5. Two worked scenario templates

These are the only two families of short question he named. Practise these two, not everything.

Template A — Pedigree / genetic counselling

Step 1: Describe the pedigree pattern before you name it. Count affected individuals per generation, note the sex of affected individuals, and check for male→male transmission.

Pattern you see	Consistent with
Every generation, both sexes, male→male transmission	Autosomal dominant
Skipped generations, siblings affected, parents unaffected	Autosomal recessive
Mostly males, no male→male transmission	X-linked recessive
Affected mother → all children; affected father → none	Mitochondrial
Affected father → all sons only	Y-linked

Step 2: Compute the risk for the specific couple. - Autosomal recessive, both obligate carriers: **1 in 4 per pregnancy** - Autosomal dominant, heterozygous parent: **1 in 2 per pregnancy** - X-linked recessive, carrier mother: **1 in 2 of sons affected, 1 in 2 of daughters carriers** - Mitochondrial, affected mother: **all children of an affected mother; no children of an affected father**

Step 3: Say what you would do next. Confirm by genetic testing; consider preimplantation genetic testing if the couple wishes to avoid transmission; offer counselling on the condition itself.

Step 4: Answer in full sentences, in English, keeping gene and condition names in English.

Template B — Survival data / Kaplan–Meier

Step 1: Read the axes. x = time in months; y = probability of survival or of progression-free survival. The curve starts at 1.0.

Step 2: Identify the groups. Which curve is which treatment, genotype, or biomarker status?

Step 3: Look for separation and its timing. - Curves overlap until some month, then diverge → the effect appears late - Curves diverge from the start → the effect is present early - The steeper early drop = worse prognosis in that group

Step 4: Read the numbers. Median survival (where the curve crosses 0.5); the number of patients per group; the log-rank p-value if shown.

Step 5: Connect to a clinical decision. Which biomarker would you test? Which treatment would you choose? Does the curve support a predictive or a prognostic claim?

Step 6: State the caveat. Tick marks are censored patients — people lost to follow-up or still event-free at the cut-off. Censoring makes the estimate conservative; the true curve could be worse.

Step 7: Answer in full sentences, in English.

---

6. Common traps to avoid

Trap	Why it loses marks	Do this instead
Answering in Chinese	Zero marks — he said he did this last year	Write English, even simple sentences
Bullet points only	"A page of bullet points is not an answer"	Full sentences that argue a case
Naming only "lactose" for the lac operon	Misses the second condition	Always say *lactose present* ***and*** *glucose low*
Treating every cell as the same	Mitochondrial disease can be nuclear too	Check the pedigree, then ask which gene
Saying "BCA is the pathway for UV damage"	BCA is base excision	Name NER for bulky lesions, BER for single bases
Writing "A and B" on an MCQ	Single best option only	Pick one; use elimination
Spending 5 minutes on one MCQ	You have 20 marks of floor to protect	90 seconds each, flag and move on
Forgetting the time unit	A curve without its axis is meaningless	Always state x and y units first
Not stating a risk per pregnancy	"50% risk" is ambiguous	Say "per pregnancy"

---

7. Seven-day plan to the midterm

Day	Task	Time
1	Read §1–§2 (exam format, the four writing rules). Write the format on one card.	30 min
2	Learn the **lac operon** logic (§3.5) and the **NHEJ → SCID** chain (§3.3) until you can say both aloud	45 min

Day	Task	Time
3	Drill MCQs Q1–Q24 under time (90 s each). Mark every one you got wrong.	45 min
4	Drill MCQs Q25–Q48 under time. Re-do the ones you got wrong *without* looking at the answer.	45 min
5	Drill MCQs Q49–Q70. Then re-do every wrong one.	45 min
6	Practise the two scenario templates (§5) by writing out two full short answers in English, in sentences, under 22 minutes each.	60 min
7	Re-read §3.1 (Kaplan–Meier), §3.2 (genetic counselling), §3.4 (mitochondrial pedigree). Sleep 8 hours.	30 min

Total: about 5 hours. That is enough, provided you do the short answers in full sentences.

---

8. Final self-test

Tick each box only if you could do it **today, unaided**.

- ☐ I can state the exam format: 20 MCQ + 2 of 3 short, 30 marks, 90 minutes, English only.
- ☐ I know the midterm is 30% of the final grade.
- ☐ I can explain in English, in full sentences, why NHEJ failure causes SCID.
- ☐ I can name both conditions for strong lac transcription and the molecule that does each job.
- ☐ I can read a Kaplan–Meier curve and say whether the gene is involved early or late.
- ☐ I can identify mitochondrial inheritance in a pedigree **and** explain why a nuclear mutation can mimic it.
- ☐ I can compute a per-pregnancy risk from a pedigree and say what test to recommend.
- ☐ I can distinguish BER from NER from MMR from HR from NHEJ by the lesion each one handles.
- ☐ I have written at least two short answers in full English sentences.
- ☐ I can name the five trp operon genes and the three lac operon genes.
- ☐ I know the optimal IPTG concentration and why CHO cells need codon optimisation.
- ☐ I know a cis element from a trans factor, with the chromosome 1 / chromosome 13 example.

If you ticked fewer than 10, you are not ready — go back to the days you have not done.

Everything above must be answered in English. Write full sentences. Defend your reasoning.

GNBF5020 · Introduction to Molecular Biology and Genetics · Dr. Tony Wong · Lectures 1–5, 12 Sep – 17 Oct 2026